

Verification Document

A Laboratory Test of Flow Trading

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1 Main Paper Tables

1.1 Table 1: Summary Statistics — Traders' Behavior (T11–T20)

None	CDA	Flow30	Flow60	CDA	Flow30	Flow60
	T1 - T20	T1 - T20	T1 - T20	T11 - T20	T11 - T20	T11 - T20
Number of Orders	31.65	23.81	25.94	31.86	23.04	24.82
Seller	16.86	12.07	12.71	17.1	11.36	12.32
Buyer	14.79	11.74	13.23	14.76	11.68	12.5
Shares per Order	114.39	186.24	192.08	111.88	188.44	194.87
Seller	114.51	184.89	190.69	111.2	185.46	196.69
Buyer	114.27	187.59	193.47	112.56	191.42	193.06
Shares per Order (early)	150.49	246.17	259.3	148.23	259.27	263.06
Seller	149.71	251.61	267.33	143.16	262.58	271.89
Buyer	152.03	240.72	250.94	153.3	255.97	253.67
Order Price	8.8	[6.72, 12.59]	[6.91, 11.79]	8.53	[6.42, 12.3]	[6.95, 11.68]
Seller	10.7	[9.21, 15.46]	[9.43, 14.08]	10.27	[8.66, 14.88]	[9.29, 13.57]
Buyer	6.9	[4.22, 9.71]	[4.39, 9.49]	6.79	[4.19, 9.71]	[4.62, 9.79]
Order Price (early)	8.98	[6.63, 12.96]	[6.67, 11.99]	8.62	[6.45, 12.7]	[6.65, 11.66]
Seller	10.77	[8.49, 15.62]	[8.59, 13.77]	10.51	[8.1, 15.04]	[8.49, 13.21]
Buyer	7.21	[4.78, 10.31]	[4.81, 10.28]	6.73	[4.8, 10.36]	[4.94, 10.24]
Order Max Rate		19.91	34.11		21.83	36.83
Seller		19.67	35.82		21.29	39.3
Buyer		20.15	32.4		22.37	34.36
Order Max Rate (early)		21.14	37.83		22.65	39.34
Seller		21.02	38.4		21.77	42.16
Buyer		21.27	37.07		23.53	36.24
Orders at Max Quantity	0.26			0.26		
Seller	0.25			0.25		
Buyer	0.27			0.26		
Orders at Speed Limit		0.38	0.22		0.48	0.29
Seller		0.38	0.23		0.46	0.3
Buyer		0.39	0.21		0.49	0.27

1.2 Table 1: Summary Statistics — Traders' Behavior (T1–T20)

None	CDA	Flow30	Flow60	CDA	Flow30	Flow60	CDA
	T1 - T20	T1 - T20	T1 - T20	T1 - T10	T1 - T10	T1 - T10	T11
Number of Orders	31.65	23.81	25.94	31.44	24.58	27.06	31.8
Seller	16.86	12.07	12.71	16.62	12.78	13.1	17.1
Buyer	14.79	11.74	13.23	14.82	11.8	13.96	14.7
Shares per Order	114.39	186.24	192.08	116.9	184.04	189.29	111.
Seller	114.51	184.89	190.69	117.82	184.32	184.7	111.
Buyer	114.27	187.59	193.47	115.97	183.75	193.89	112.
Order Price	8.8	[6.72, 12.59]	[6.91, 11.79]	9.07	[7.01, 12.88]	[6.86, 11.89]	8.53
Seller	10.7	[9.21, 15.46]	[9.43, 14.08]	11.13	[9.76, 16.04]	[9.56, 14.59]	10.2
Buyer	6.9	[4.22, 9.71]	[4.39, 9.49]	7.01	[4.26, 9.71]	[4.16, 9.19]	6.79
Order Max Rate		19.91	34.11		18.0	31.39	
Seller		19.67	35.82		18.06	32.33	
Buyer		20.15	32.4		17.93	30.45	
Orders at Max Quantity	0.26			0.27			0.26
Seller	0.25			0.26			0.25
Buyer	0.27			0.28			0.26
Orders at Speed Limit		0.38	0.22		0.29	0.15	
Seller		0.38	0.23		0.29	0.16	
Buyer		0.39	0.21		0.28	0.14	

1.3 Table 2: Summary Statistics — Market Performance (T11–T20)

None	CDA	Flow30	Flow60	CDA	P
	T1 - T20	T1 - T20	T1 - T20	T11 - T20	T
Clearing Price	8.62	9.37	9.45	8.21	9
Std. Dev.	(2.77)	(2.54)	(2.14)	(2.51)	(
$P_t - P_{t-1}$	0.57	0.39	0.43	0.55	0
Std. Dev. of $(\ln P_t - \ln P_{t-1})$	0.14	0.07	0.08	0.13	0
PPI	3.98	0.89	0.78	3.88	0
Traded Volume (shares)	1138.36	959.6	1121.25	1179.1	1
Traded Vol. / Q_{CE} (%)	93.61	78.75	92.07	96.9	8
Filled Contract / Q_{CE} (%)	85.15	77.12	88.05	89.26	8
Num. of Transactions	14			15	
Mean Transaction Size	78			74	
Std. Dev.	(60.72)			(60.35)	
Clearing Rate		9.42	9.69		9
Std. Dev.		(2.61)	(2.42)		(
Time with no Flow (%)		10.74	11.97		9
Info. Effic.: $P - P_{CE}$	2.16	2.08	2.18	2.14	1
Alloc. Effic.: P / P_{CE} (x100)	84.3	81.81	90.04	88.85	8
$P^{buy} / P^{buy}_{CE} - P^{sell} / P^{sell}_{CE}$	53.7	11.18	9.97	80.03	2

1.4 Table 2: Summary Statistics — Market Performance (T1–T20)

None	CDA	Flow30	Flow60	CDA	Flow60
	T1 - T20	T1 - T20	T1 - T20	T1 - T10	T1 - T20
Clearing Price	8.62	9.37	9.45	9.04	9.5
Std. Dev.	(2.77)	(2.54)	(2.14)	(2.95)	(2.5)
$P_t - P_{t-1}$	0.57	0.39	0.43	0.58	0.3
Std. Dev. of $(\ln P_t - \ln P_{t-1})$	0.14	0.07	0.08	0.15	0.0
PPI	3.98	0.89	0.78	4.09	1.0
Traded Volume (shares)	1138.36	959.6	1121.25	1097.62	87
Traded Vol. / Q_{CE} (%)	93.61	78.75	92.07	90.32	71
Filled Contract / Q_{CE} (%)	85.15	77.12	88.05	81.03	69
Num. of Transactions	14			13	
Mean Transaction Size	78			82	
Std. Dev.	(60.72)			(60.91)	
Clearing Rate		9.42	9.69		9.5
Std. Dev.		(2.61)	(2.42)		(2.5)
Time with no Flow (%)		10.74	11.97		11
Info. Effic.: $P_t - P_{CE}$	2.16	2.08	2.18	2.19	2.3
Alloc. Effic.: P_t / P_{CE} (x100)	84.3	81.81	90.04	79.76	77
$P^{buy}_t / P^{buy}_{CE} - P^{sell}_t / P^{sell}_{CE}$	53.7	11.18	9.97	27.38	-5

1.5 Table 3: Regression — Trader Behavior Metrics

	(1)	(2)	(3)	(4)
Intercept	36.222*** (4.385)	41.674*** (5.946)	154.056*** (26.272)	160.10*** (46.27)
block_1[T.True]	-3.475 (3.982)	0.000*** (0.000)	-42.140* (23.071)	-0.000 (0.00)
block_2[T.True]	-1.577 (2.873)	0.000** (0.000)	-20.502 (17.503)	0.00 (0.00)
block_3[T.True]	-1.329 (1.976)	-3.056 (2.371)	-16.841 (11.972)	-19.6 (17.74)
block_4[T.True]	-0.098 (1.252)	-1.110 (1.544)	-5.038 (8.602)	-7.06 (11.60)
format[T.Flow30]	-7.840*** (0.787)	-8.820*** (0.791)	71.849*** (3.815)	76.558*** (5.86)
format[T.Flow60]	-5.710*** (0.835)	-7.040*** (0.930)	77.693*** (5.282)	82.989*** (7.61)
period	-0.312 (0.231)	-0.565* (0.308)	-2.168 (1.433)	-2.67 (2.51)
CDA avg. Flow30 = Flow60? (p)	31.6 0.007	31.9 0.038	114.4 0.315	111. 0.44
Observations	300	150	300	150
R^2	0.343	0.505	0.613	0.63
Adjusted R^2	0.327	0.487	0.604	0.61
Residual Std. Error	4.777 (df=292)	3.932 (df=144)	28.738 (df=292)	29.507 (d
F Statistic	15.542*** (df=7; 292)	27.029*** (df=5; 144)	70.104*** (df=7; 292)	51.631*** (d

Note:

*p<0.1; **p<0.05; ***p<0.01

1.6 Table 4: Regression — Price Volatility and Liquidity

	(1)	(2)	(3)	(4)
Intercept	2.200*** (0.314)	2.522*** (0.460)	0.324*** (0.058)	0.36*** (0.058)
block_1[T.True]	-0.717*** (0.270)	-0.000*** (0.000)	-0.151*** (0.051)	0.00*** (0.000)
block_2[T.True]	-0.756*** (0.202)	0.000*** (0.000)	-0.111*** (0.037)	0.00*** (0.000)
block_3[T.True]	-0.624*** (0.144)	-0.753*** (0.181)	-0.094*** (0.027)	-0.10*** (0.000)
block_4[T.True]	-0.464*** (0.084)	-0.572*** (0.111)	-0.051*** (0.016)	-0.00*** (0.000)
format[T.Flow30]	-0.744*** (0.062)	-0.561*** (0.072)	-0.070*** (0.010)	-0.00*** (0.000)
format[T.Flow60]	-0.684*** (0.065)	-0.470*** (0.077)	-0.059*** (0.010)	-0.00*** (0.000)
interval	0.002 (0.003)	0.004 (0.003)		
period	-0.051*** (0.016)	-0.079*** (0.022)	-0.010*** (0.003)	-0.00*** (0.000)
CDA avg.	0.57	0.55	0.14	0.00
Flow30 = Flow60? (p)	0.013	0.012	0.101	0.20
Observations	6900	3450	300	18
R^2	0.114	0.083	0.283	0.20
Adjusted R^2	0.113	0.081	0.266	0.20
Residual Std. Error	0.954 (df=6891)	0.896 (df=3443)	0.054 (df=292)	0.052 (df=17)
F Statistic	23.215*** (df=8; 6891)	13.254*** (df=6; 3443)	11.039*** (df=7; 292)	6.406*** (df=7; 17)

Note:

1.7 Table 5: Regression — Volume

	(1)	(2)	(3)	(4)
Intercept	639.702*** (163.943)	664.099** (283.195)	0.552*** (0.121)	0.595* (0.166)
block_1[T.True]	267.794* (144.312)	0.000*** (0.000)	0.186* (0.106)	0.000 (0.000)
block_2[T.True]	262.853** (114.041)	0.000*** (0.000)	0.130 (0.083)	0.000* (0.000)
block_3[T.True]	430.013*** (76.200)	400.604*** (112.682)	0.070 (0.056)	0.041 (0.069)
block_4[T.True]	223.038*** (52.019)	213.453*** (74.364)	0.094*** (0.034)	0.082 (0.043)
format[T.Flow30]	-178.764*** (28.710)	-132.314*** (38.217)	-0.080*** (0.021)	-0.048 (0.021)
format[T.Flow60]	-17.110 (27.064)	-3.756 (39.827)	0.029 (0.020)	0.020 (0.022)
period	24.945*** (8.789)	22.548 (15.356)	0.019*** (0.006)	0.017 (0.009)
CDA avg. Flow30 = Flow60? (p)	1138.4 < 0.001	1179.1 < 0.001	0.85 < 0.001	0.89 < 0.001
Observations	300	150	300	150
R^2	0.488	0.347	0.296	0.199
Adjusted R^2	0.476	0.324	0.279	0.171
Residual Std. Error	158.365 (df=292)	157.341 (df=144)	0.119 (df=292)	0.100 (df=144)
F Statistic	34.186*** (df=7; 292)	14.876*** (df=5; 144)	14.183*** (df=7; 292)	7.936*** (df=5; 144)

Note:

*p<0.1; **p<0.05; ***p<0.01

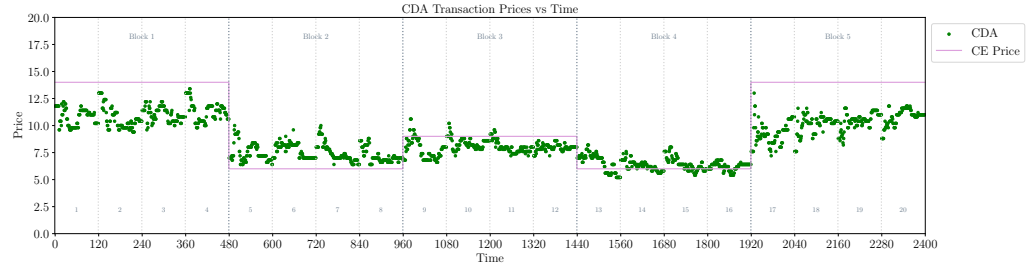
1.8 Table 6: Regression — Efficiency and Buyer-Seller Disparity

	(1)	(2)	(3)	
Intercept	8.592*** (0.527)	8.980*** (0.816)	0.537*** (0.094)	0.069*** (0.005)
block_1[T.True]	-4.494*** (0.455)	0.000*** (0.000)	0.157* (0.081)	0.069*** (0.005)
block_2[T.True]	-4.210*** (0.342)	0.000** (0.000)	0.183*** (0.062)	0.069*** (0.005)
block_3[T.True]	-5.044*** (0.235)	-5.001*** (0.315)	0.121*** (0.042)	0.069*** (0.005)
block_4[T.True]	-2.845*** (0.148)	-2.910*** (0.195)	0.095*** (0.024)	0.069*** (0.005)
format[T.Flow30]	-0.006 (0.087)	-0.025 (0.110)	-0.025 (0.017)	-0.006 (0.087)
format[T.Flow60]	0.162* (0.085)	0.247** (0.115)	0.057*** (0.014)	0.162* (0.085)
interval	-0.046*** (0.005)	-0.054*** (0.007)		-0.046*** (0.005)
period	-0.252*** (0.028)	-0.271*** (0.044)	0.019*** (0.005)	-0.252*** (0.028)
CDA avg. Flow30 = Flow60? (p)	2.16 0.019	2.14 0.003	0.843 < 0.001	2.16 0.019
Observations	7200	3600	300	7200
R^2	0.372	0.482	0.378	0.372
Adjusted R^2	0.371	0.482	0.363	0.371
Residual Std. Error	1.701 (df=7191)	1.470 (df=3593)	0.089 (df=292)	1.701 (df=7191)
F Statistic	234.841*** (df=8; 7191)	203.046*** (df=6; 3593)	16.060*** (df=7; 292)	234.841*** (df=8; 7191)

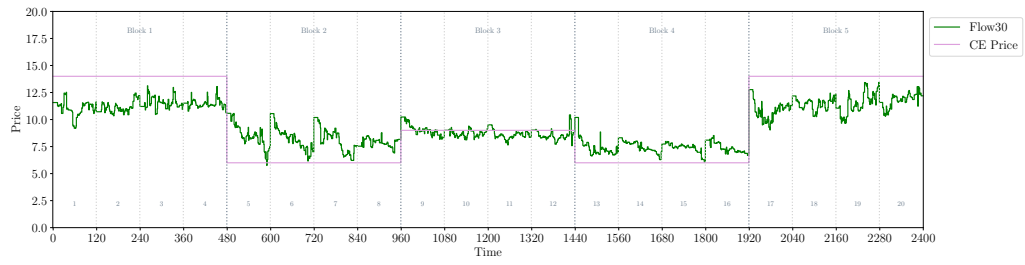
Note:

2 Main Paper Figures

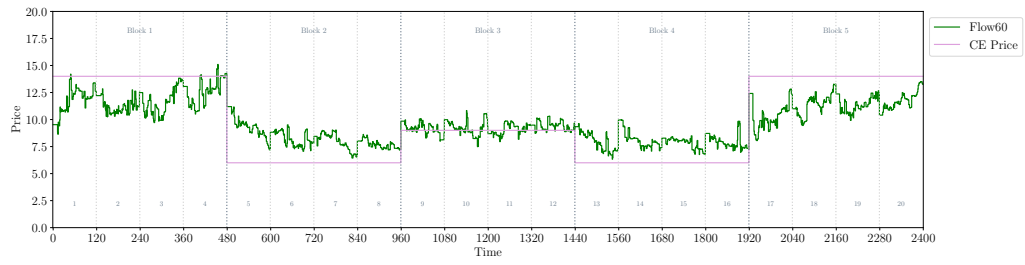
2.1 Figure 4a: CDA Transaction Prices



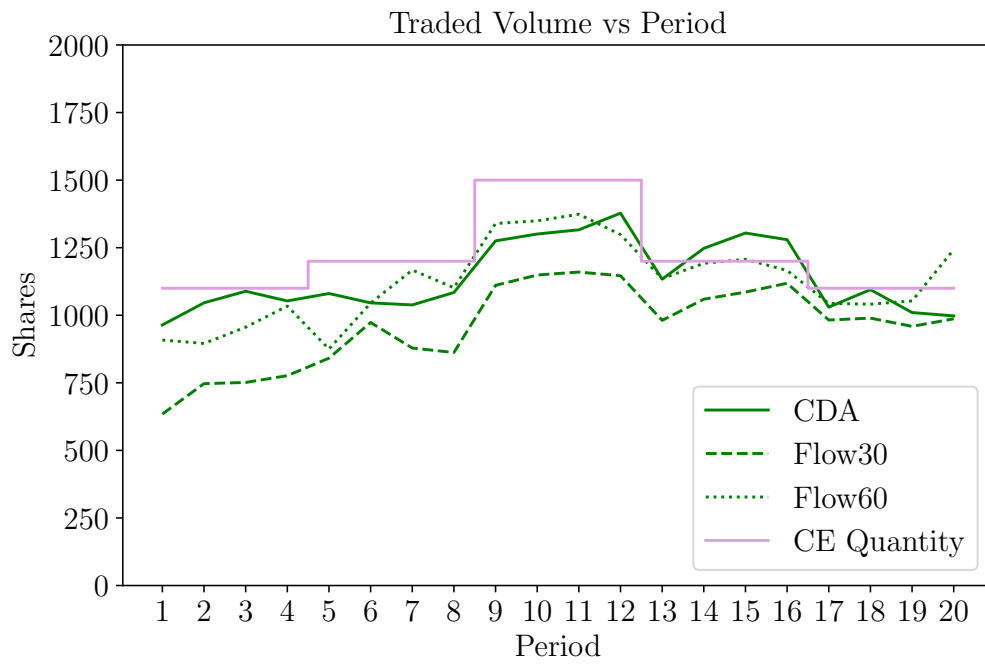
2.2 Figure 4b: Flow30 Transaction Prices



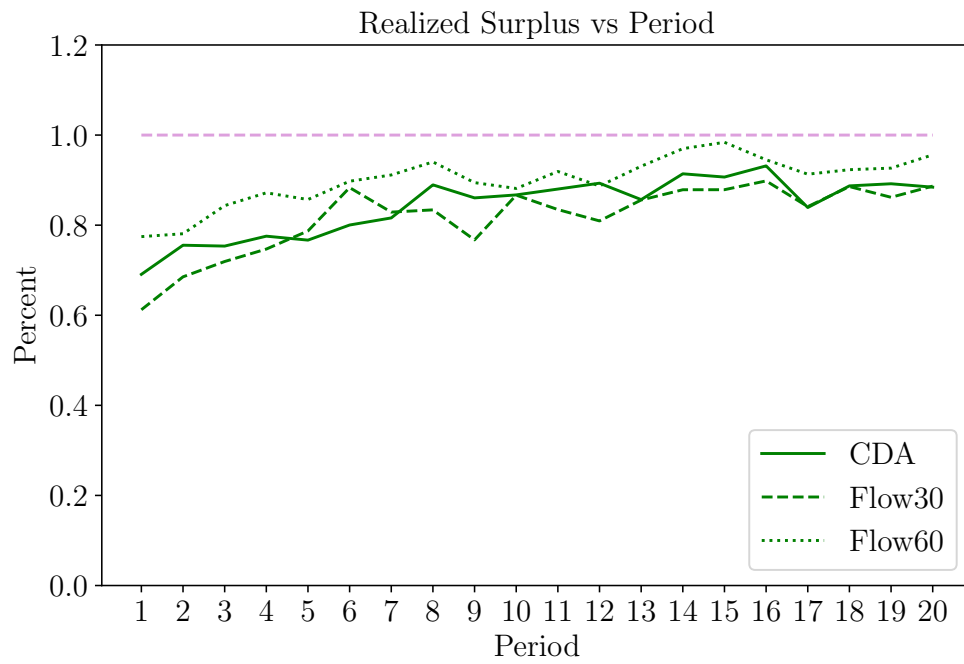
2.3 Figure 4c: Flow60 Transaction Prices



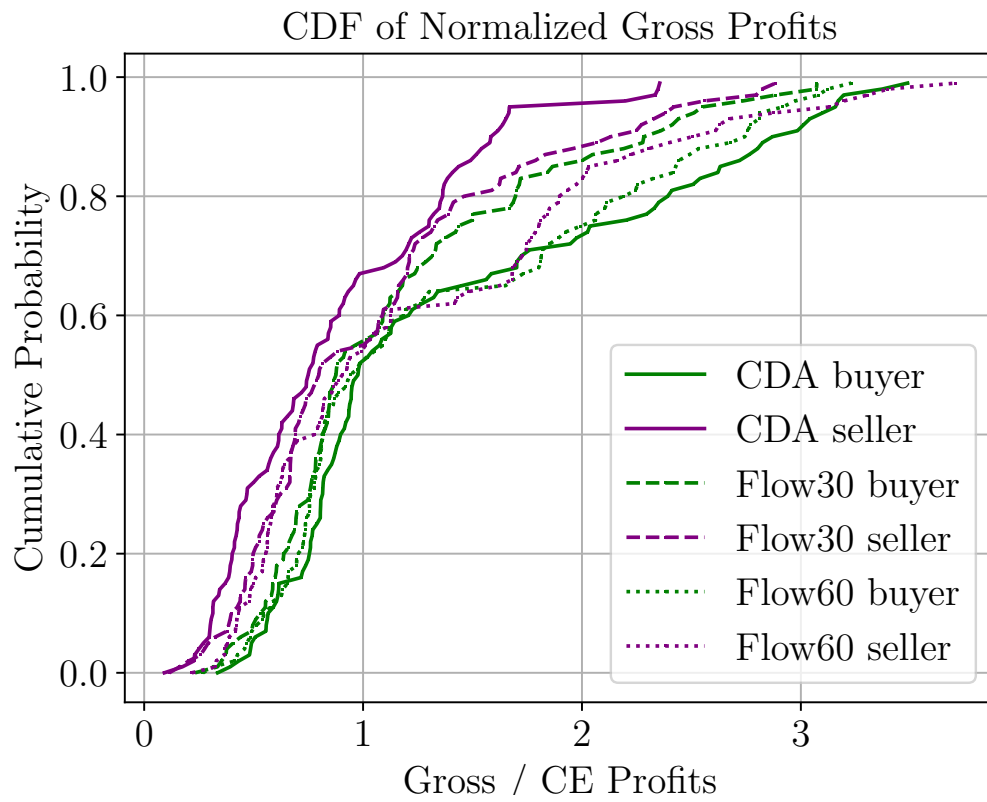
2.4 Figure 5a: Trade Volume vs. Period



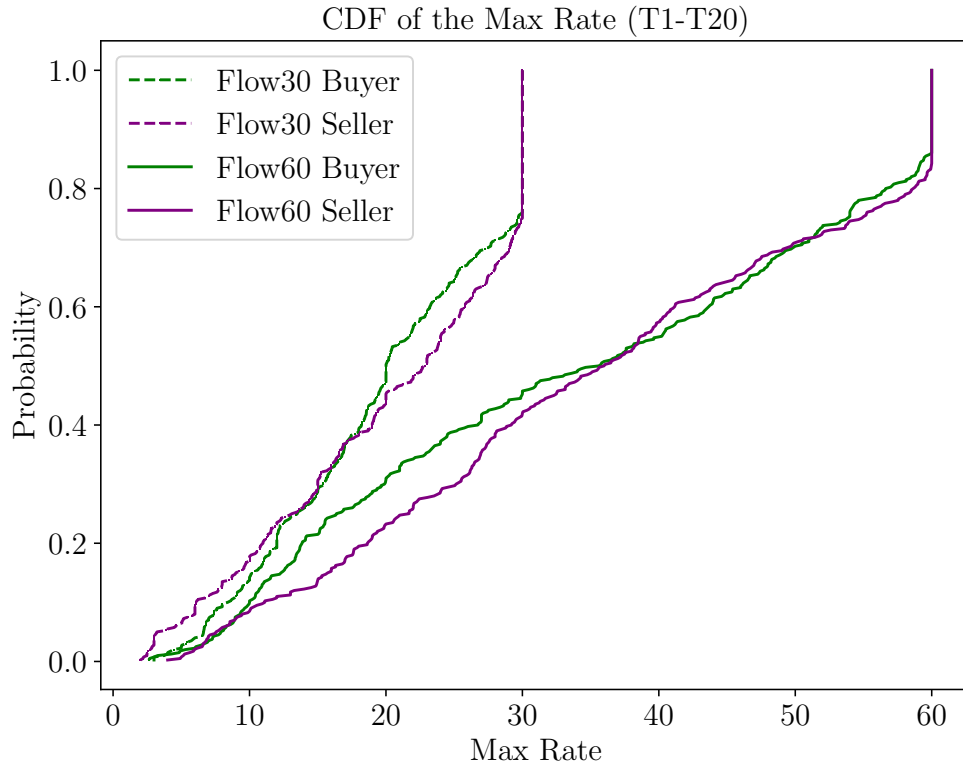
2.5 Figure 5b: Realized Surplus vs. Period



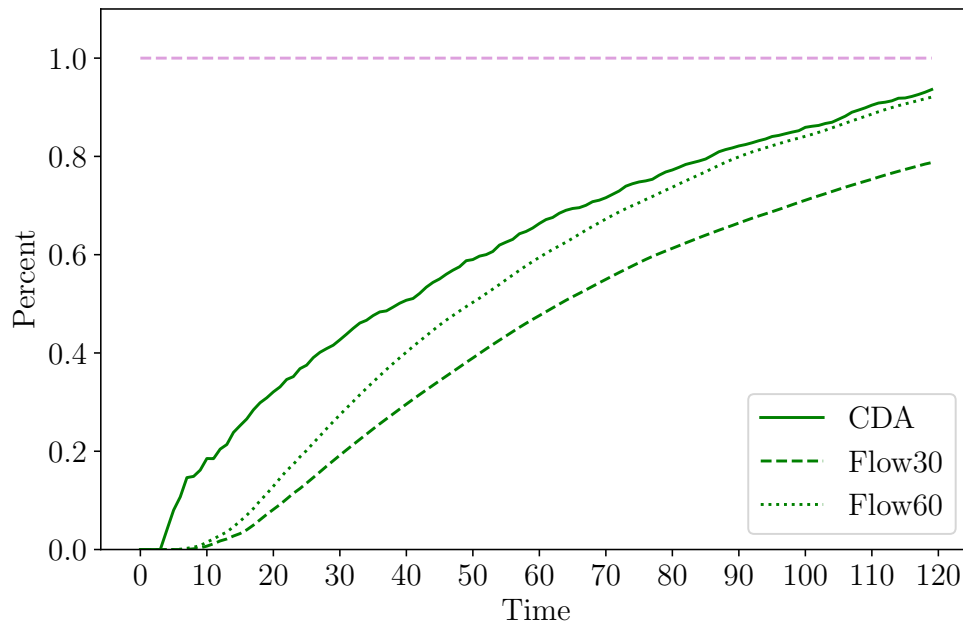
2.6 Figure 6: Profit Distribution (CDF)



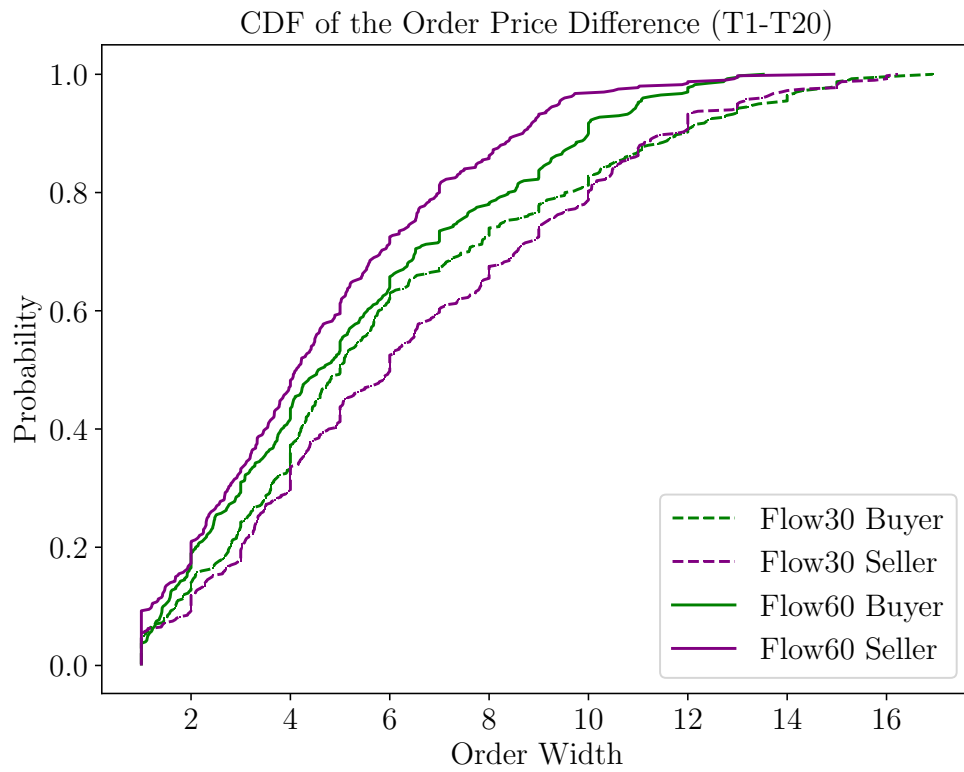
2.7 Figure 7: CDF of U_{max}



2.8 Figure 8: Cumulative Executed Volume



2.9 Figure 9: CDF of Price-Range Width



3 Supplemental Appendix Tables

3.1 Table S1: Spearman Correlations

Table 1: Spearman Rank Correlations Between Behavioral Variables (Flow Traders)

	Flow30 buy T1-T10	Flow30 buy T11-T20	Flow30 sell T1-T10	Flow30 sell T11-T20	Flow60 buy T1-T10	Flow60 buy T11-T20	Flow60 sell T1-T10	Flow60 sell T11-T20
Excess Profit $\times$ Price Markup	0.017	-0.074	-0.230	-0.026	0.045	0.244	0.040	-0.056
Excess Profit $\times U_{max}$ (%)	-0.243	-0.180	-0.044	-0.066	0.089	0.109	0.214	-0.066
Excess Profit $\times$ Price Range	-0.000	0.193	0.059	-0.053	0.183	0.254	-0.040	0.138
Price Markup $\times U_{max}$ (%)	-0.132	0.026	-0.106	-0.150	0.187	0.199	0.091	0.238
Price Markup $\times$ Price Range	0.088	0.089	0.117	0.039	0.258	0.113	0.019	-0.208
U_{max} (%) $\times$ Price Range	-0.370	-0.359	0.314	0.278	0.244	0.121	-0.243	-0.257

3.2 Table S2: Market Performance (10s intervals)

None	CDA	Flow30	Flow60	CDA	P
	T1 - T20	T1 - T20	T1 - T20	T11 - T20	T
Clearing Price	8.62	9.37	9.45	8.21	9
Std. Dev.	(2.77)	(2.54)	(2.14)	(2.51)	(
$P_t - P_{t-1}$	0.57	0.39	0.43	0.55	0
Std. Dev. of $(\ln P_t - \ln P_{t-1})$	0.14	0.07	0.08	0.13	0
PPI	3.98	0.89	0.78	3.88	0
Traded Volume (shares)	1138.36	959.6	1121.25	1179.1	1
Traded Vol. / Q_{CE} (%)	93.61	78.75	92.07	96.9	8
Filled Contract / Q_{CE} (%)	85.15	77.12	88.05	89.26	8
Num. of Transactions	14			15	
Mean Transaction Size	78			74	
Std. Dev.	(60.72)			(60.35)	
Clearing Rate		9.42	9.69		9
Std. Dev.		(2.61)	(2.42)		(
Time with no Flow (%)		10.74	11.97		9
Info. Effic.: $P - P_{CE}$	2.16	2.08	2.18	2.14	1
Alloc. Effic.: P / P_{CE} (x100)	84.3	81.81	90.04	88.85	8
$P^{buy} / P^{buy}_{CE} - P^{sell} / P^{sell}_{CE}$	53.7	11.18	9.97	80.03	2

3.3 Table S3: Price Volatility Regression (10s)

	(1)	(2)	(3)	(4)
Intercept	2.292*** (0.335)	2.595*** (0.542)	0.388*** (0.063)	0.45*** (0.11)
block_1[T.True]	-0.963*** (0.275)	0.000*** (0.000)	-0.194*** (0.056)	0.00*** (0.00)
block_2[T.True]	-0.861*** (0.211)	0.000*** (0.000)	-0.128*** (0.041)	0.00*** (0.00)
block_3[T.True]	-0.734*** (0.153)	-0.808*** (0.212)	-0.113*** (0.029)	-0.13*** (0.00)
block_4[T.True]	-0.568*** (0.091)	-0.644*** (0.129)	-0.064*** (0.019)	-0.07*** (0.00)
format[T.Flow30]	-0.695*** (0.066)	-0.627*** (0.087)	-0.085*** (0.011)	-0.07*** (0.00)
format[T.Flow60]	-0.636*** (0.068)	-0.552*** (0.093)	-0.067*** (0.012)	-0.05*** (0.00)
interval	0.018*** (0.006)	0.017** (0.008)		
period	-0.057*** (0.016)	-0.076*** (0.027)	-0.012*** (0.003)	-0.03*** (0.00)
CDA avg.	0.85	0.81	0.17	0.17
Flow30 = Flow60? (p)	0.059	0.103	0.027	0.103
Observations	3300	1650	300	1650
R^2	0.121	0.116	0.288	0.288
Adjusted R^2	0.119	0.113	0.271	0.271
Residual Std. Error	0.959 (df=3291)	0.937 (df=1643)	0.065 (df=292)	0.061 (df=292)
F Statistic	19.777*** (df=8; 3291)	11.639*** (df=6; 1643)	11.814*** (df=7; 292)	6.248*** (df=7; 292)

Note:

3.4 Table S4: Price Deviation Regression (10s)

Table 2: Regression: Price Deviation from Competitive Equilibrium

	$ P_\tau - P_{CE} $	
	T1-T20 (1)	T11-T20 (2)
Intercept	8.571*** (0.655)	8.871*** (1.023)
Flow30	0.022 (0.104)	0.013 (0.134)
Flow60	0.186* (0.100)	0.273** (0.138)
Period number	-0.252*** (0.035)	-0.265*** (0.055)
Interval number	-0.094*** (0.012)	-0.110*** (0.016)
Observations	3600	1800
Adjusted R^2	0.385	0.497
CDA avg.	2.14	2.08

3.5 Table S5: Market Performance (2s intervals)

None	CDA	Flow30	Flow60	CDA	P
	T1 - T20	T1 - T20	T1 - T20	T11 - T20	T
Clearing Price	8.62	9.37	9.45	8.21	9
Std. Dev.	(2.77)	(2.54)	(2.14)	(2.51)	(
$P_t - P_{t-1}$	0.57	0.39	0.43	0.55	0
Std. Dev. of $(\ln P_t - \ln P_{t-1})$	0.14	0.07	0.08	0.13	0
PPI	3.98	0.89	0.78	3.88	0
Traded Volume (shares)	1138.36	959.6	1121.25	1179.1	1
Traded Vol. / Q_{CE} (%)	93.61	78.75	92.07	96.9	8
Filled Contract / Q_{CE} (%)	85.15	77.12	88.05	89.26	8
Num. of Transactions	14			15	
Mean Transaction Size	78			74	
Std. Dev.	(60.72)			(60.35)	
Clearing Rate		9.42	9.69		9
Std. Dev.		(2.61)	(2.42)		(
Time with no Flow (%)		10.74	11.97		9
Info. Effic.: $P - P_{CE}$	2.16	2.08	2.18	2.14	1
Alloc. Effic.: P / P_{CE} (x100)	84.3	81.81	90.04	88.85	8
$P^{buy} / P^{buy}_{CE} - P^{sell} / P^{sell}_{CE}$	53.7	11.18	9.97	80.03	2

3.6 Table S6: Price Volatility Regression (2s)

	(1)	(2)	(3)	(4)
Intercept	2.293*** (0.297)	2.565*** (0.408)	0.220*** (0.046)	0.260*** (0.046)
block_1[T.True]	-0.687*** (0.241)	-0.000*** (0.000)	-0.100** (0.041)	0.000 (0.000)
block_2[T.True]	-0.732*** (0.182)	0.000*** (0.000)	-0.075** (0.029)	0.000 (0.000)
block_3[T.True]	-0.583*** (0.128)	-0.724*** (0.157)	-0.061*** (0.021)	-0.075*** (0.021)
block_4[T.True]	-0.457*** (0.077)	-0.552*** (0.098)	-0.035*** (0.013)	-0.045*** (0.013)
format[T.Flow30]	-0.991*** (0.060)	-0.817*** (0.072)	-0.046*** (0.007)	-0.035*** (0.007)
format[T.Flow60]	-0.904*** (0.062)	-0.718*** (0.075)	-0.033*** (0.008)	-0.025*** (0.008)
interval	-0.001 (0.001)	-0.000 (0.001)		
period	-0.049*** (0.014)	-0.073*** (0.019)	-0.006** (0.002)	-0.006** (0.002)
CDA avg.	0.27	0.26	0.10	0.10
Flow30 = Flow60? (p)	< 0.001	< 0.001	0.036	0.200
Observations	17700	8850	300	1800
R^2	0.155	0.132	0.200	0.155
Adjusted R^2	0.155	0.131	0.181	0.155
Residual Std. Error	0.927 (df=17691)	0.847 (df=8843)	0.043 (df=292)	0.041 (df=292)
F Statistic	42.924*** (df=8; 17691)	27.030*** (df=6; 8843)	7.115*** (df=7; 292)	4.204*** (df=7; 292)

Note:

3.7 Table S7: Price Deviation Regression (2s)

Table 3: Regression: Price Deviation from Competitive Equilibrium

	$ P_\tau - P_{CE} $	
	T1-T20	T11-T20
	(1)	(2)
Intercept	8.654*** (0.403)	8.976*** (0.610)
Flow30	-0.018 (0.073)	-0.030 (0.093)
Flow60	0.150** (0.073)	0.246** (0.096)
Period number	-0.255*** (0.021)	-0.271*** (0.032)
Interval number	-0.019*** (0.002)	-0.022*** (0.002)
Observations	18000	9000
Adjusted R^2	0.359	0.466
CDA avg.	2.17	2.13
Flow30 = Flow60?	0.000	0.000

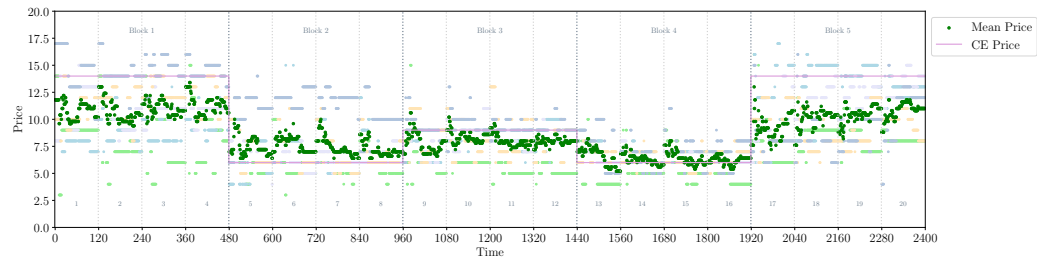
3.8 Table S8: Unweighted Price Statistics

Table 4: Summary Statistics with Unweighted Prices

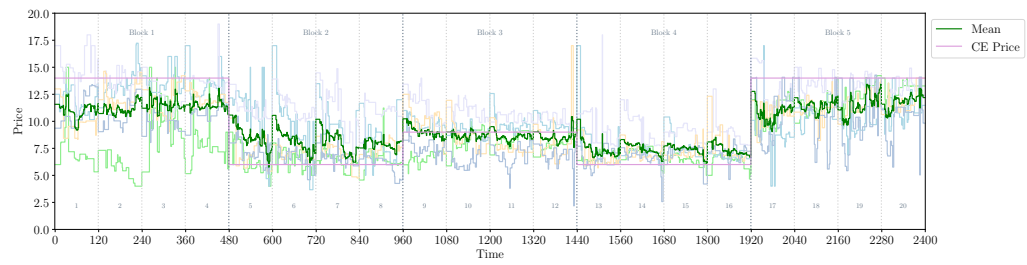
	All Periods (T1–T20)			First Half (T1–T10)			Second Half (T11–T20)		
	CDA	Flow30	Flow60	CDA	Flow30	Flow60	CDA	Flow30	Flow60
Average Price	8.59	9.43	9.71	8.97	9.59	9.88	8.21	9.26	9.53
$ P_t - P_{CE} $	-1.21	-0.37	-0.09	-0.83	-0.21	0.08	-1.59	-0.54	-0.27
$ P_t - P_{t-1} $	0.14	0.15	0.17	0.14	0.15	0.19	0.14	0.15	0.16
$\text{Std}(\ln P_t - \ln P_{t-1})$	0.08	0.05	0.06	0.08	0.05	0.07	0.07	0.05	0.06

4 Supplemental Appendix Figures

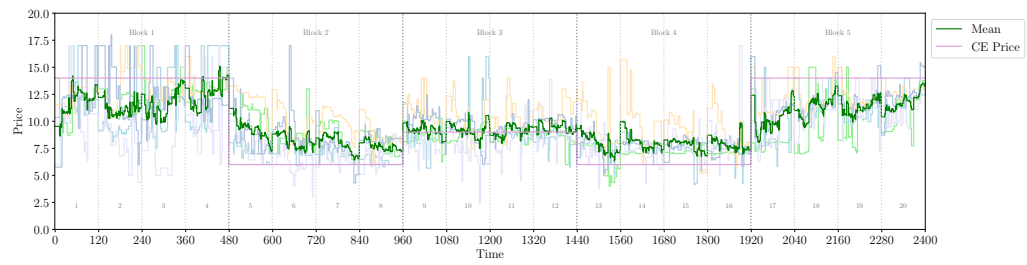
4.1 Figure S1a: CDA Prices (All Groups)



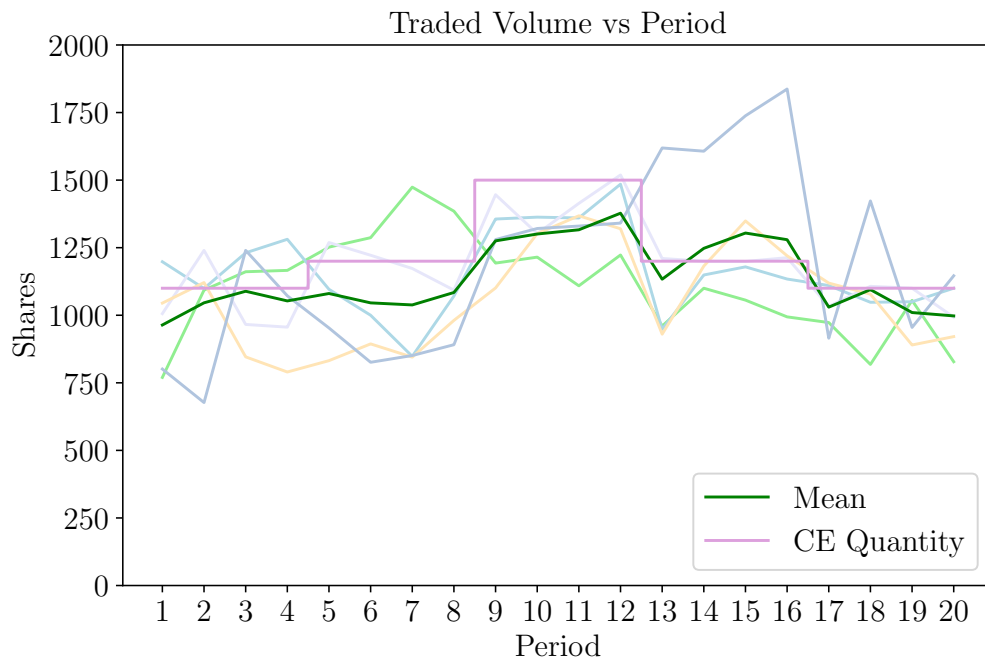
4.2 Figure S1b: Flow30 Prices (All Groups)



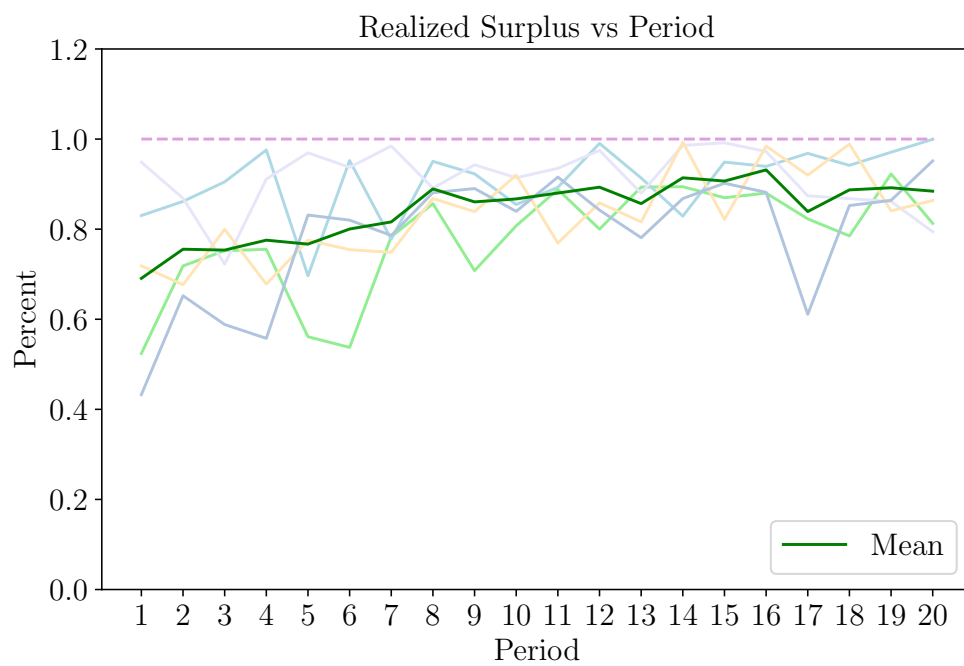
4.3 Figure S1c: Flow60 Prices (All Groups)



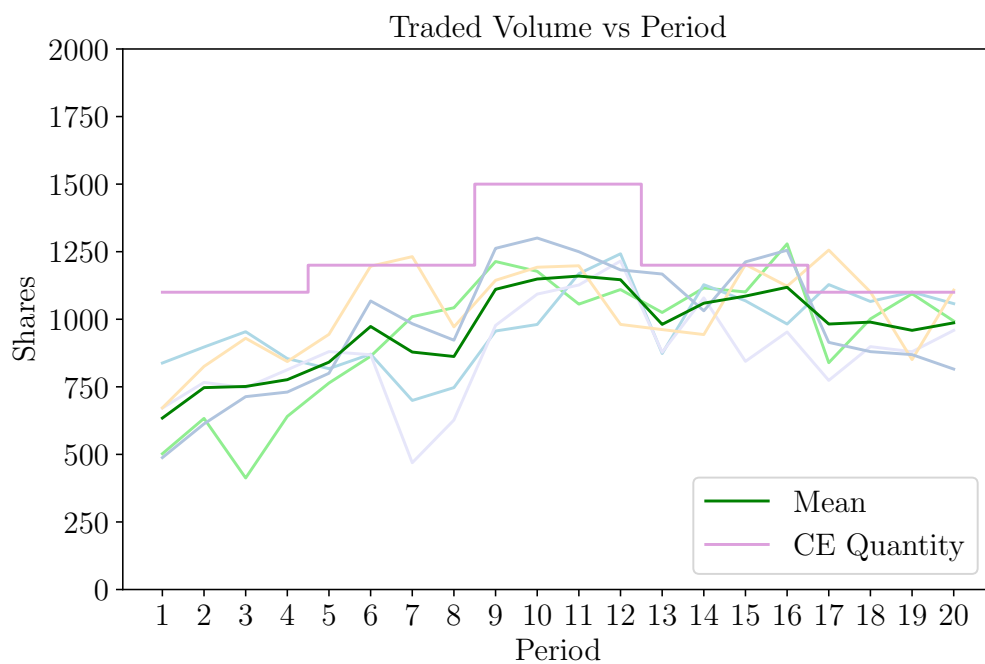
4.4 Figure S2a: CDA Trade Volume



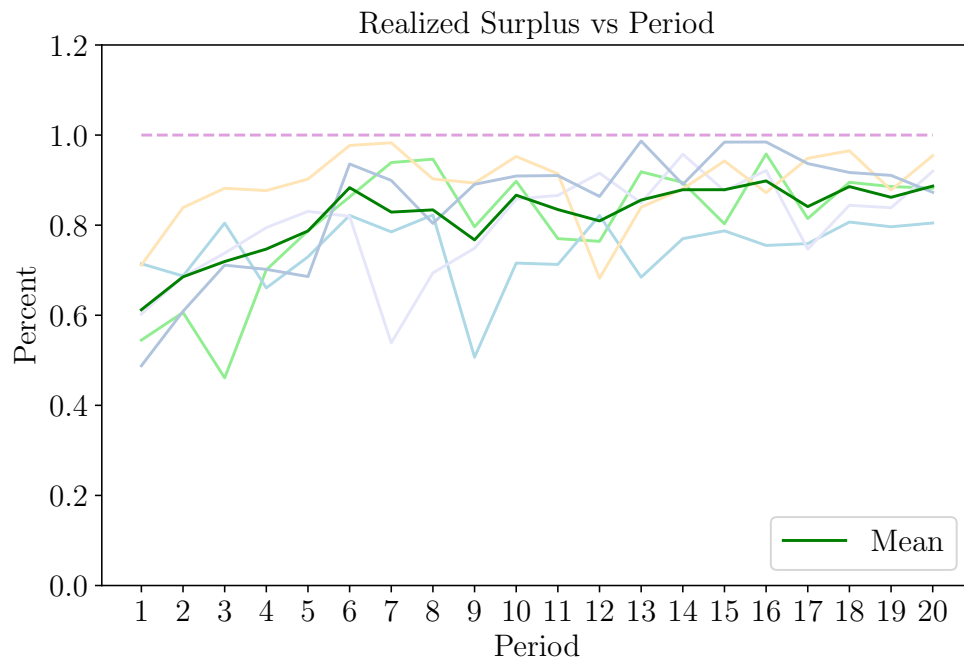
4.5 Figure S2b: CDA Realized Surplus



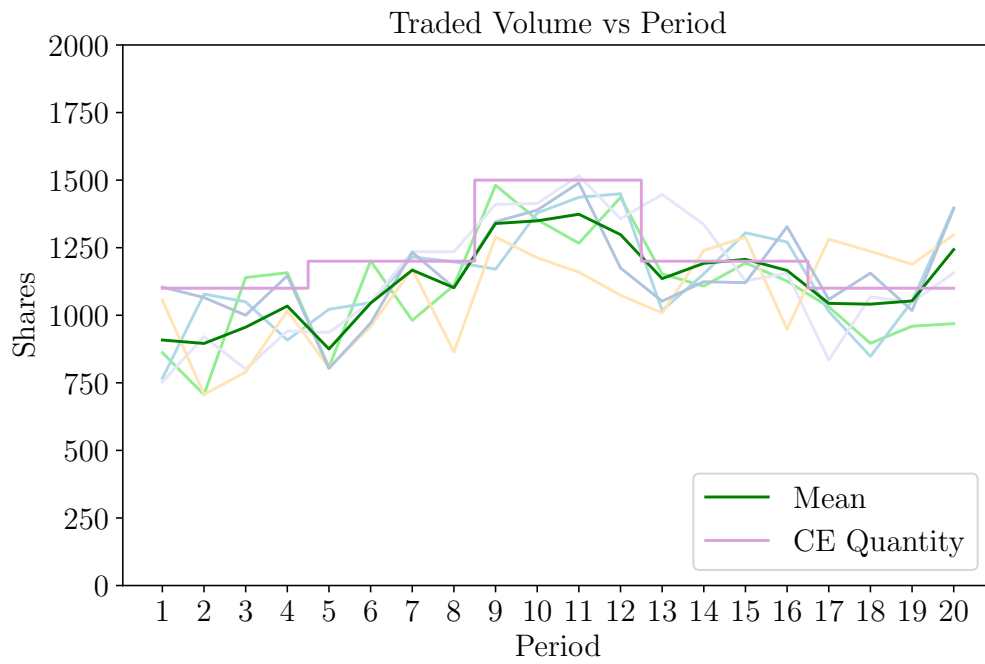
4.6 Figure S2c: Flow30 Trade Volume



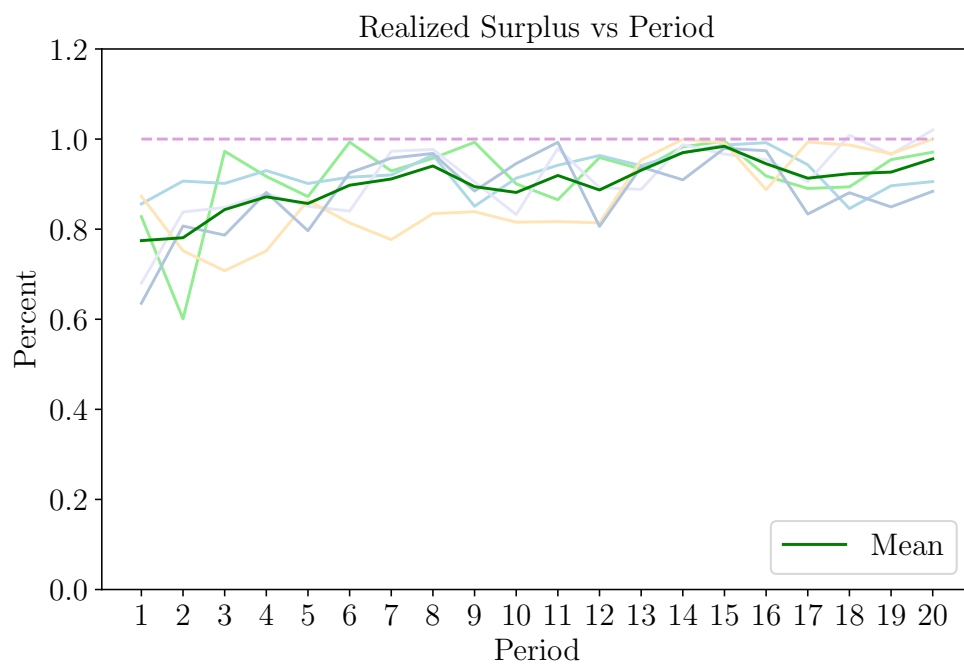
4.7 Figure S2d: Flow30 Realized Surplus



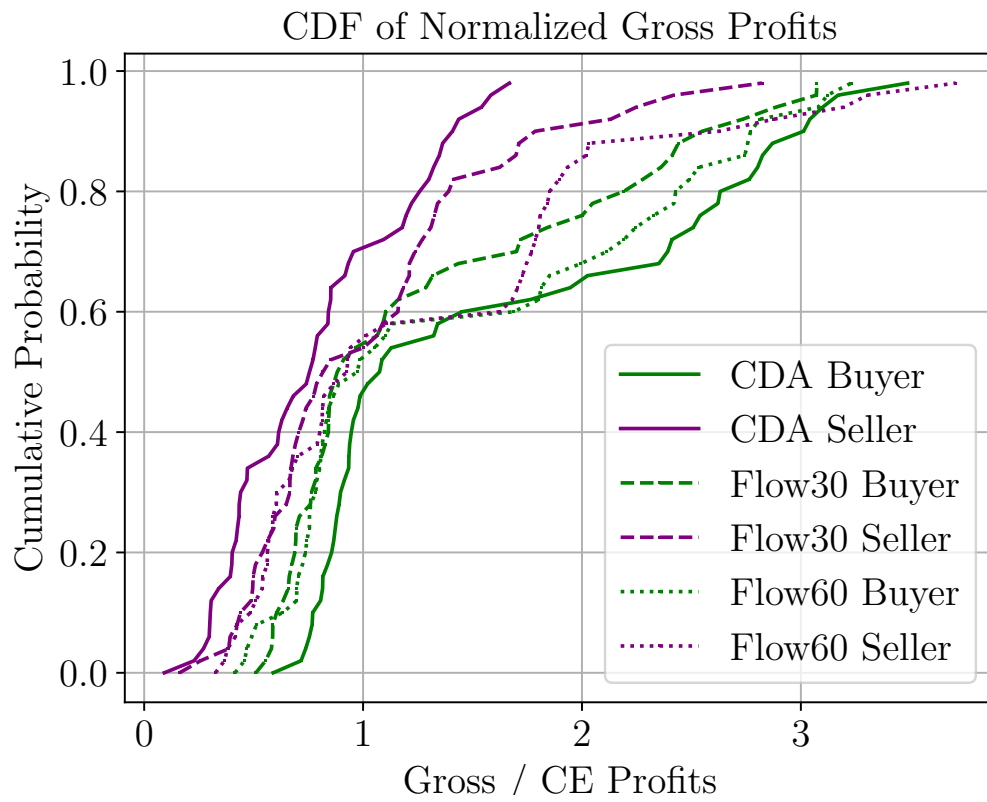
4.8 Figure S2e: Flow60 Trade Volume



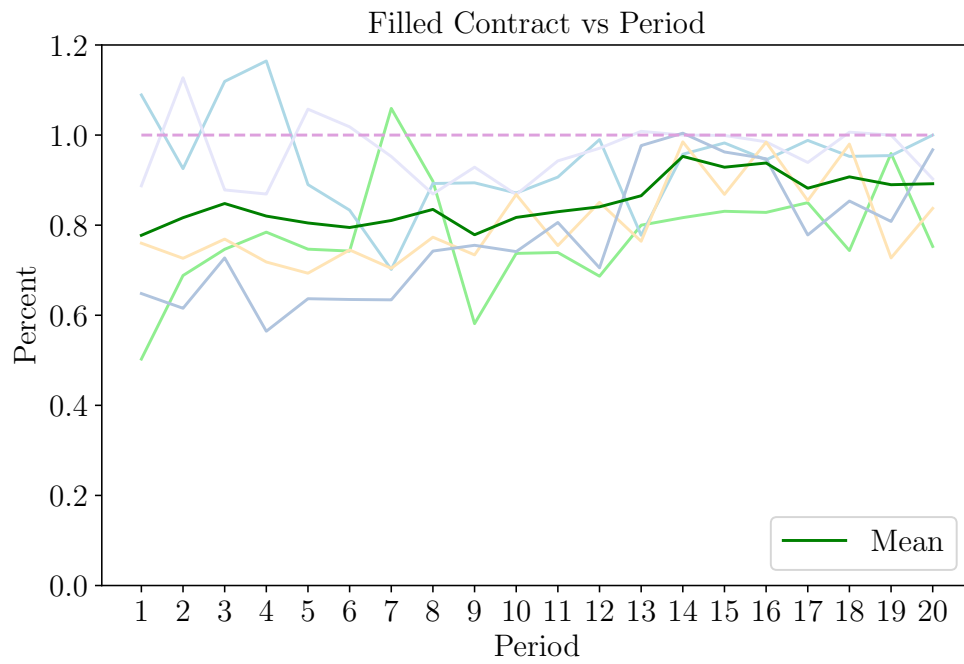
4.9 Figure S2f: Flow60 Realized Surplus



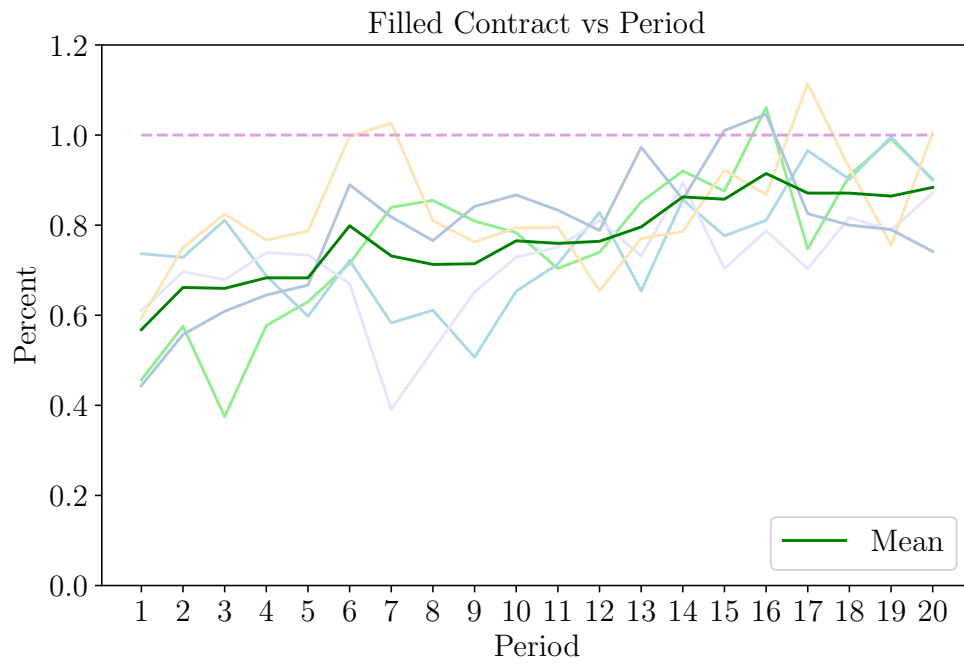
4.10 Figure S3: Profit Distribution (Last 10 Periods)



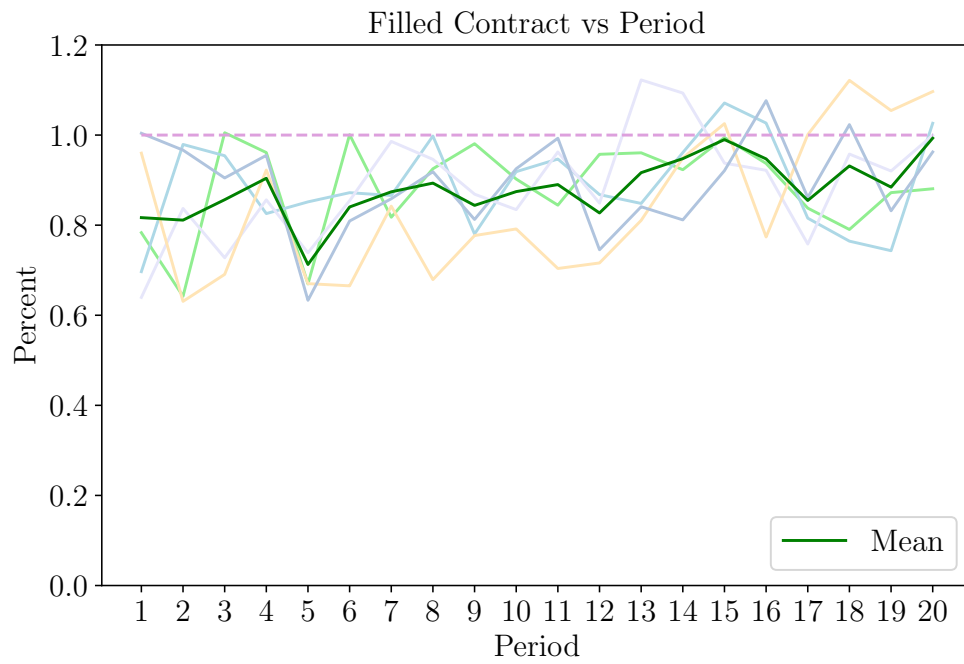
4.11 Figure S4a: CDA Contract Execution



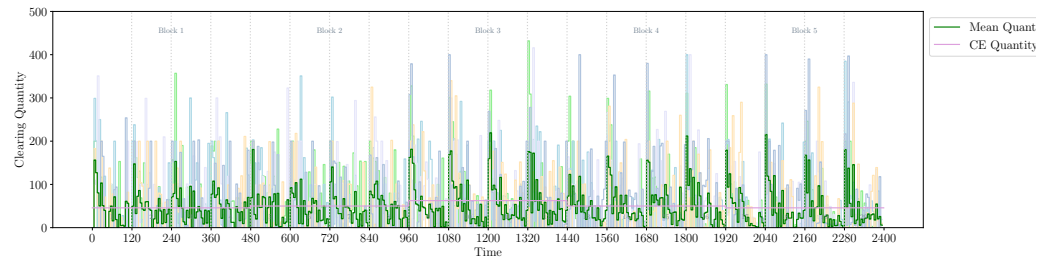
4.12 Figure S4b: Flow30 Contract Execution



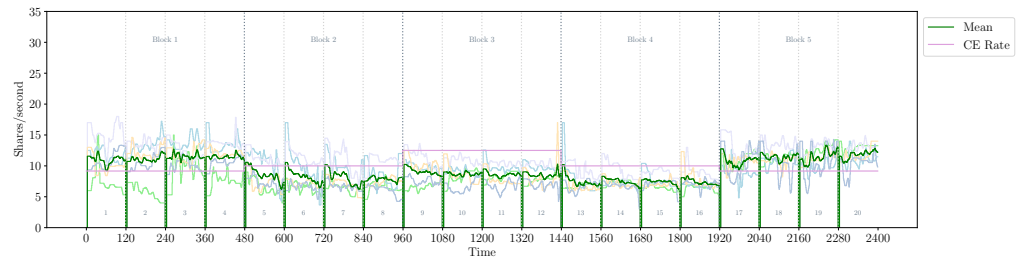
4.13 Figure S4c: Flow60 Contract Execution



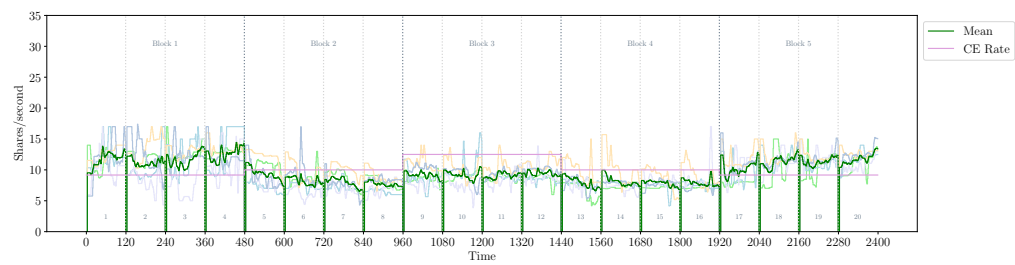
4.14 Figure S5a: CDA Execution Pace



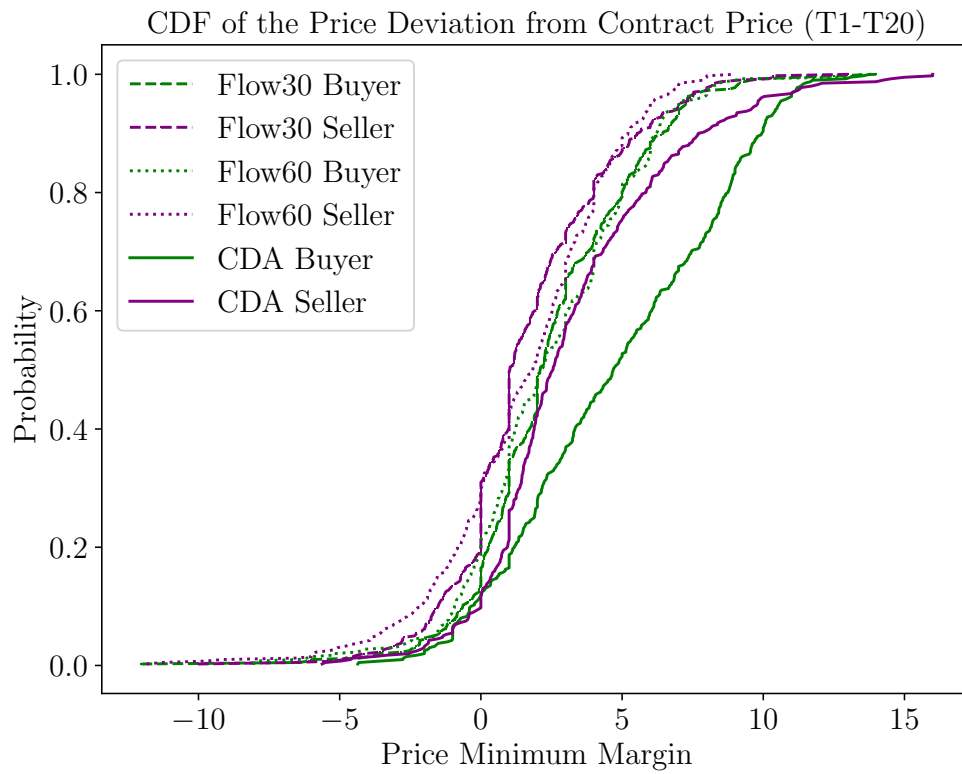
4.15 Figure S5b: Flow30 Execution Pace



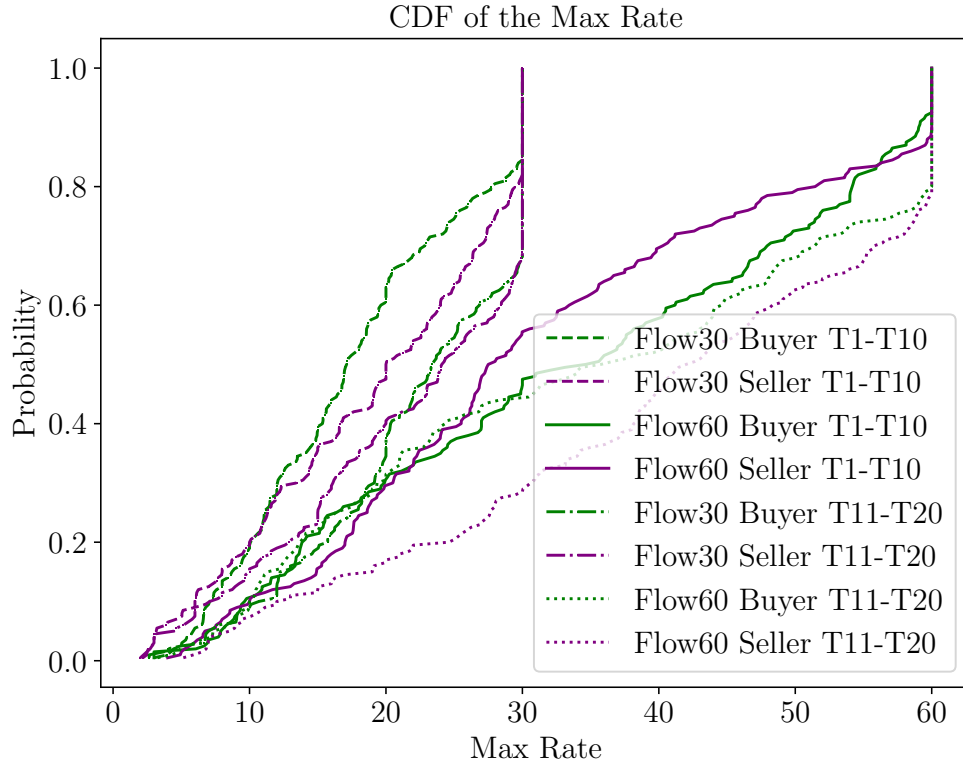
4.16 Figure S5c: Flow60 Execution Pace



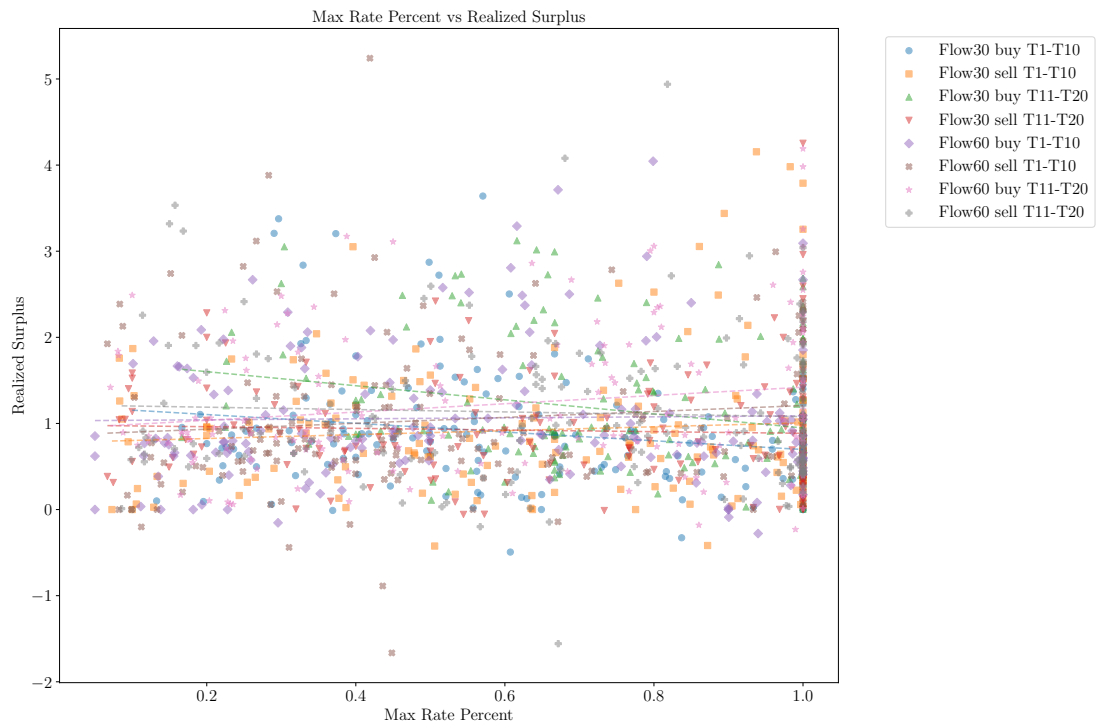
4.17 Figure S6: Price Markup CDF



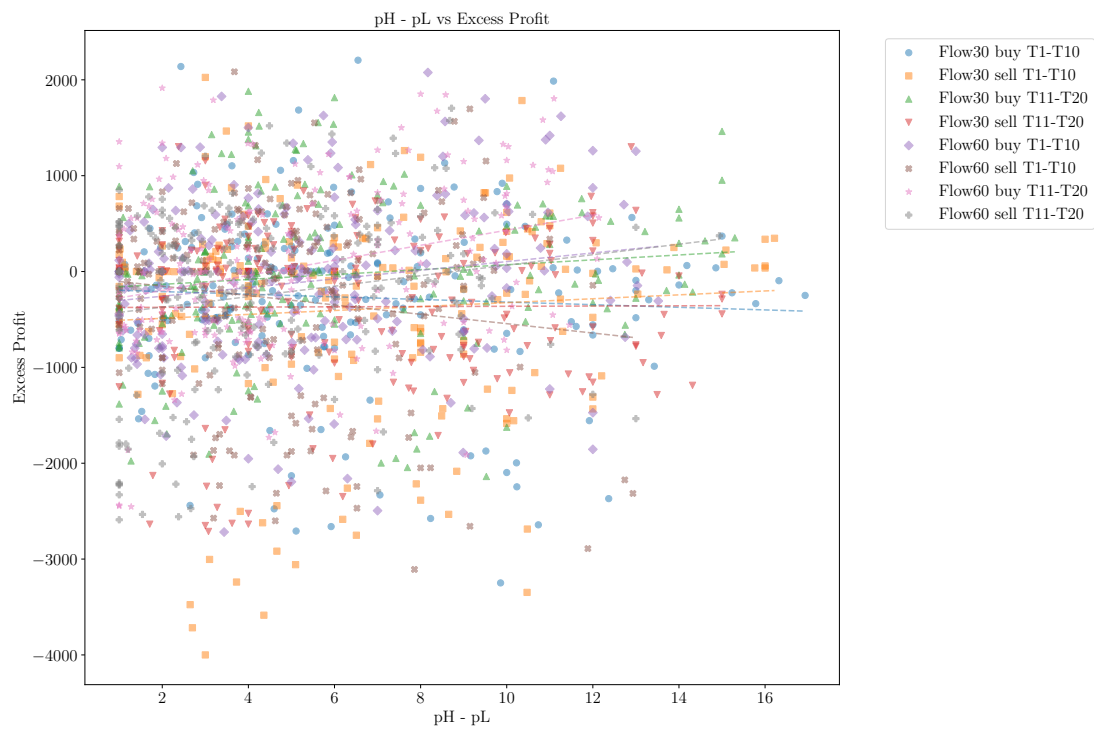
4.18 Figure S7: U_{max} CDF (Early vs. Late)



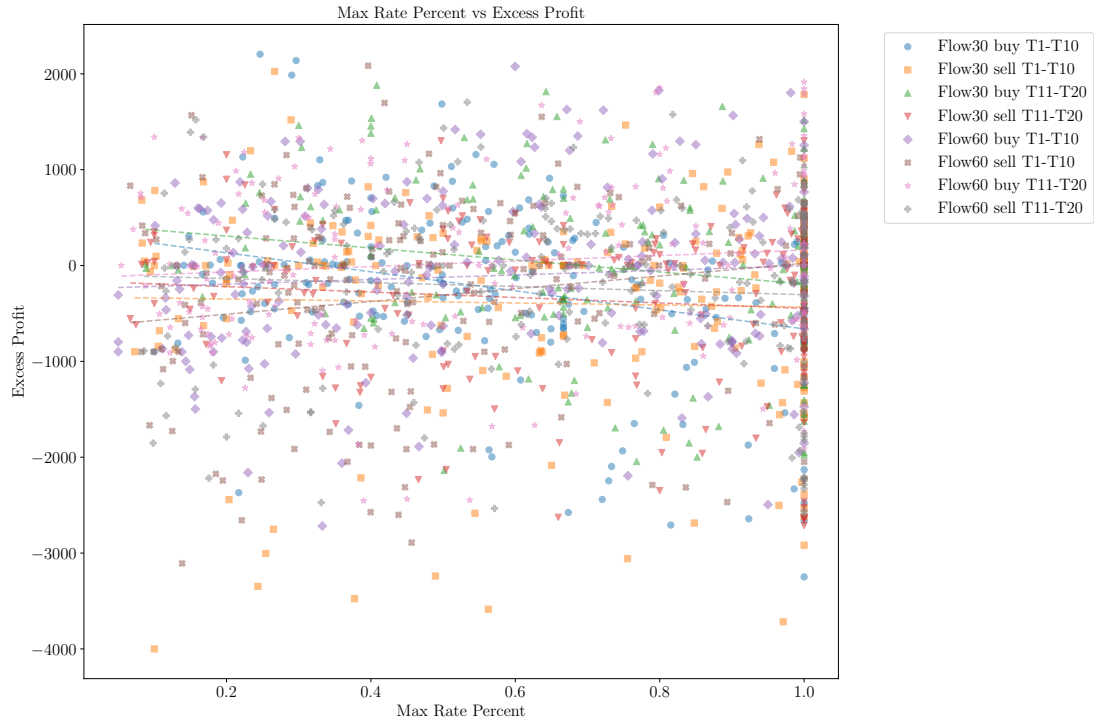
4.19 Figure S8: Speed Limit Usage vs. Surplus



4.20 Figure S9a: Price-Range vs. Excess Profit



4.21 Figure S9b: U_{max} vs. Excess Profit



4.22 Figure S9c: Price Markup vs. Excess Profit

